

BIGFRAME
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Positioning Guidance

Defect Inspection

Precision Metrology

Deep Learning

TridiVision

Universal Platform Software

2D+3D+AI

Proprietary Matching Algorithms: Fast & Highly Accurate

TridiVision Universal Software Platform

TridiVision software is a powerful graphical development platform designed specifically for machine vision engineers, offering industry-leading functionality and flexibility. It utilizes an intuitive dataflow programming architecture and includes a rich set of powerful image analysis operators, significantly improving development efficiency.

Product Description

TridiVision software is a simple and efficient vision platform that supports 2D/3D vision tasks and multi-brand cameras, allowing you to quickly build various application solutions.

The advantages are as follows:



Integrates over 270+ operators, supporting deep learning and AI rapid modeling with small samples



Meet complex and diverse application requirements in 2D and 3D vision fields, including positioning guidance, measurement, defect detection, and object recognition



Expanded toolset functionality, achieving high modularity and engineering readability



Powerful self-developed contour matching algorithm, high speed, high precision, and supports scaling matching and perspective deformation matching



HMI interface with rich control types



Support for Automatic Robot Path Planning and 3D Trajectory Generation



Deep pseudo-color point cloud acquisition, real-time preview, loop acquisition, repeatability testing readability



Provides sample programs for various typical applications, allowing for flexible development of visual application solutions



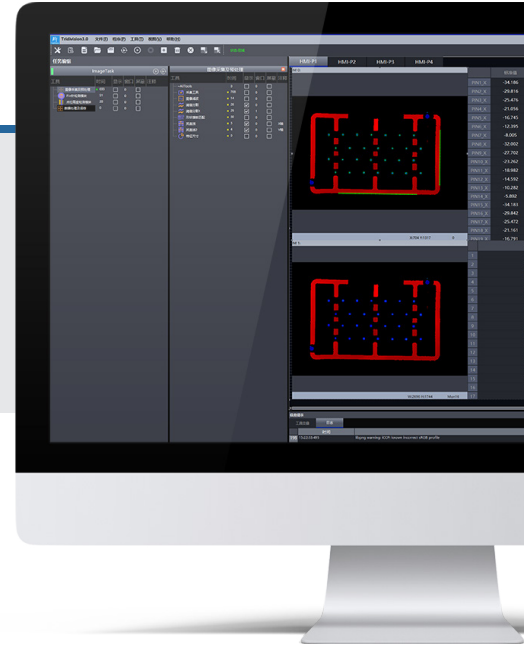
Support customized algorithm development

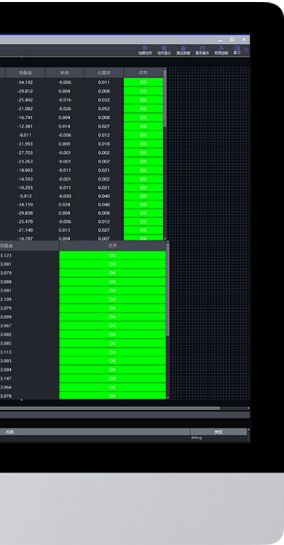


Supports numerous third-party 2D and 3D camera brands, including Hikvision, Basler, Daheng, LMI, Keyence, Smartray, and Deep Vision

Version & Function Description

Software Version	Supported Function Modules															
	1D	2D		3D		3D-zmap			Image Processing	Point Cloud Processing	Defect Detection	Deep Learning		Calibration	Operation	Communication
	Detection	Position	Detection	Identify	Path Planning	Position	Detection	Process				Training	Reasoning			
TridiVision v3.0-2D Red	✓	✓	✓						✓		✓			✓	✓	✓
TridiVision v3.0-3D-zmap(2D) Orange	✓	✓	✓			✓	✓	✓	✓		✓			✓	✓	✓
TridiVision v3.0-3D-poincloud(2D+3D-zmap) Yellow	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓
TridiVision v3.0-AI-running(2D+3D-zmap) Green	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
TridiVision v3.0-AI-training(all in one) Blue	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
HT-E5 BASIC DONGLE (USB hardware key)	/															





The unique advantage of this software lies in its deep support for professional users - whether it is quickly building standard applications or developing highly complex vision systems, TridiVision can provide strong support to help engineers achieve various machine vision tasks efficiently and accurately.

List of Features

Intuitive User Interface

1 One-click Drag and Drop

All programming operations are performed step-by-step by selecting the required operators and following the image processing logic and process, eliminating the need for tedious code writing. This allows you to focus on designing and optimizing computer vision algorithms and devote more energy to core vision tasks.

2 All Information at a Glance

Test results are visualized and displayed clearly through multiple configurable data preview windows. As program parameters change, the preview content updates in real time, allowing you to instantly observe the effects of adjustments and quickly iterate and optimize the visual algorithm.

3 HMI Designer

You can easily create a customized graphical user interface to build an entire machine vision application through a complete software package. This makes the interface design and functionality seamlessly connected, greatly improving development efficiency and user experience.

Powerful Features

1 550+ Efficient Operators

Over 550 ready-made machine vision operators, tested and optimized for hundreds of application scenarios, are available. These operators offer advanced features such as outlier suppression, sub-pixel precision processing, and arbitrary region of interest (ROI) selection. These operators meet diverse visual inspection requirements and ensure high-precision and efficient image processing.

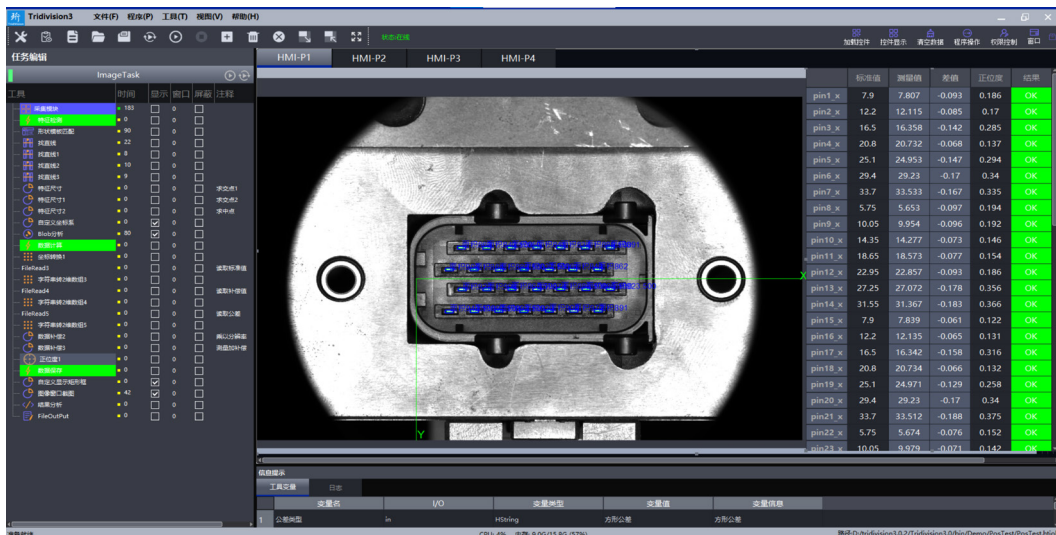
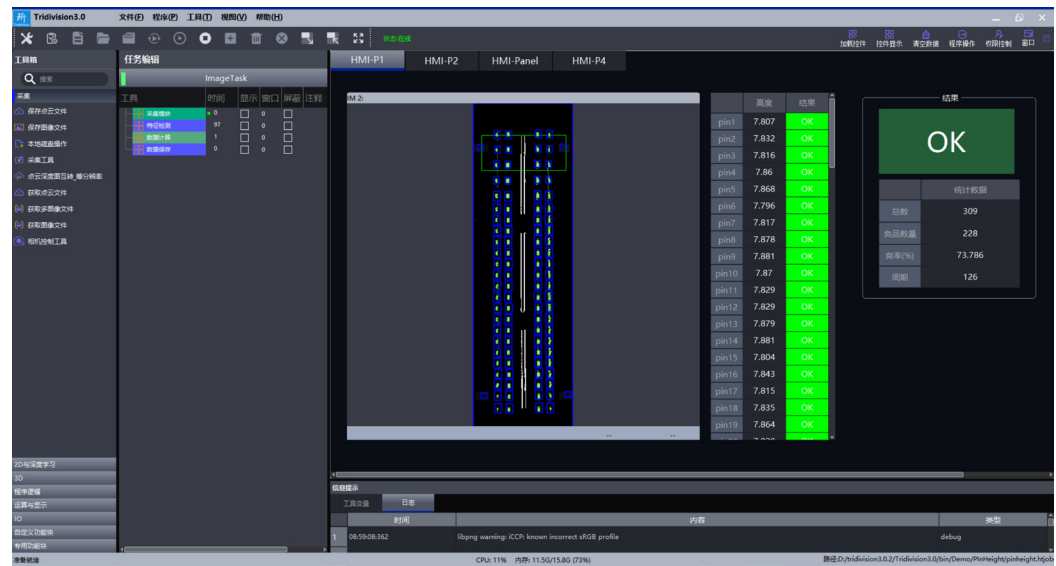
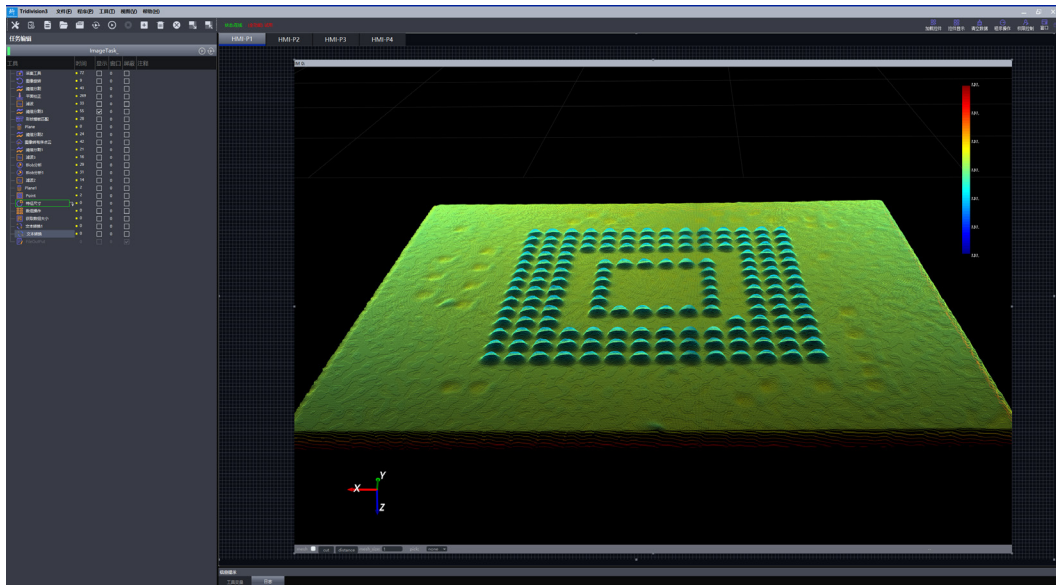
2 Hardware Acceleration

These operators are carefully optimized to fully utilize the hardware performance. Our implementation is one of the fastest in the world, ensuring excellent processing speed and responsiveness even under high load and complex computing tasks.

3 Loops & Conditions

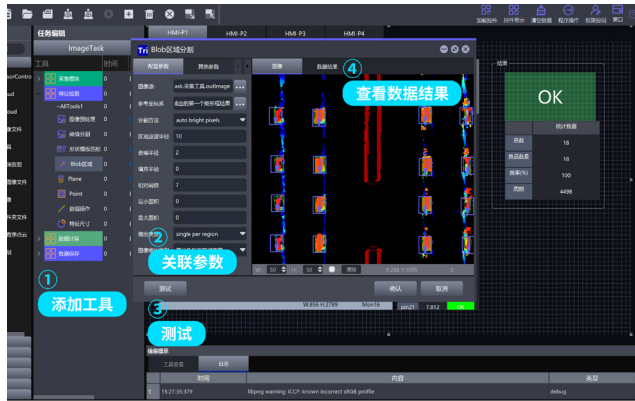
Without writing a single line of code, you can easily create custom and scalable program flows. Loops, conditionals, and subroutines are implemented graphically, making program logic design intuitive and flexible, greatly improving development efficiency and simplifying the implementation of complex processes.

Software Interface



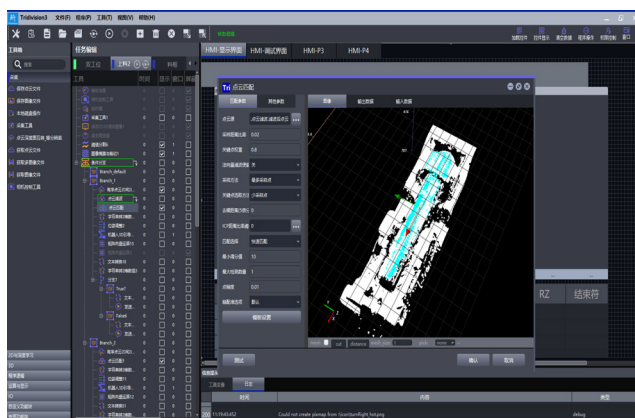
Products Introduction

Simple and Fast Graphical Programming



Add tools, associate parameters, test, view data results, and use NO CODE graphical programming for simple programming.

Rich 3D Detection Algorithms



The powerful 3D module supports height measurement, multi-camera stitching, and object recognition with grasping guidance in 3D environments.

Complete Calibration Module



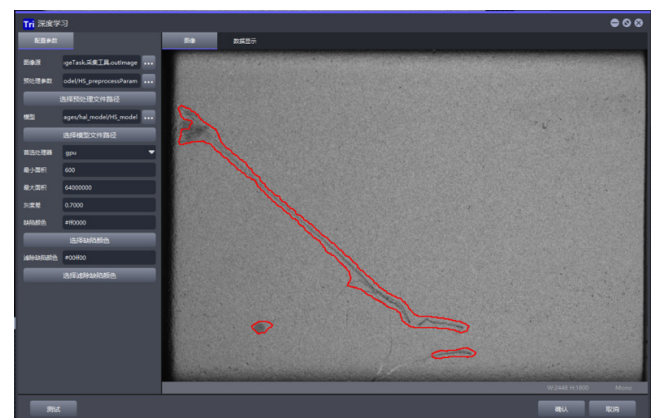
The software has a built-in automatic calibration module, under the guidance of which users can quickly deploy automatic hand-eye calibration in the field of 2D or 3D robot grasping.

Flexible Human-computer Interaction Interface



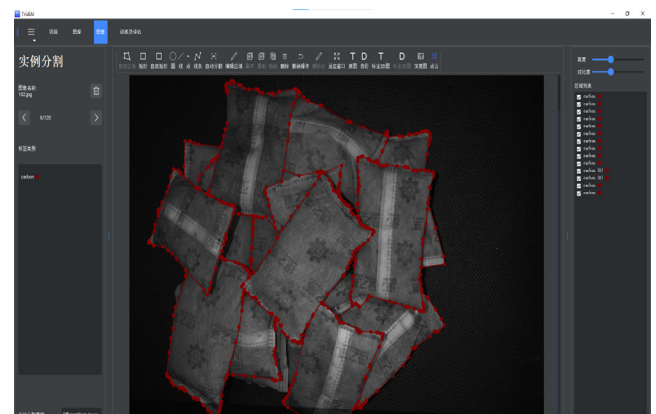
The HMI designer can flexibly drag and drop display controls and input controls to form a beautiful user interaction interface according to user application requirements.

Advanced Deep Learning Algorithms



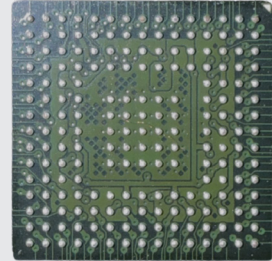
The software's built-in deep learning module supports deep learning classification, semantic segmentation, object recognition and other applications.

AI Small Sample Rapid Modeling



Intelligent simple annotation, small number of image samples, short model training time, support for model optimization and improvement.

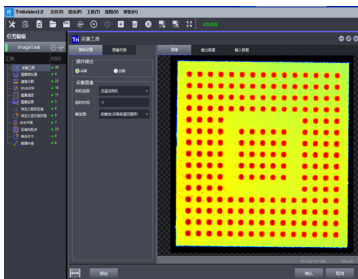
Sample Program——3D - Solder Ball Height Measurement



1

Collection Tools

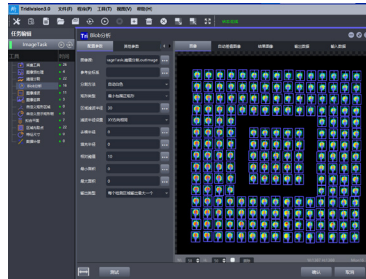
—Connect a 3D camera to capture images of the object being measured.



2

Threshold Segmentation → Blob Analysis

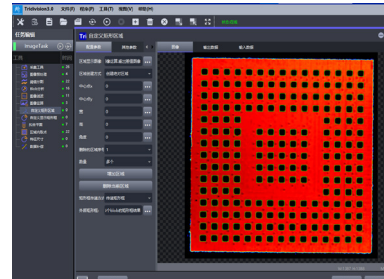
—Get the detection feature of multi-area solder ball position.



3

Filter → Image Operation → Custom Rectangular Area

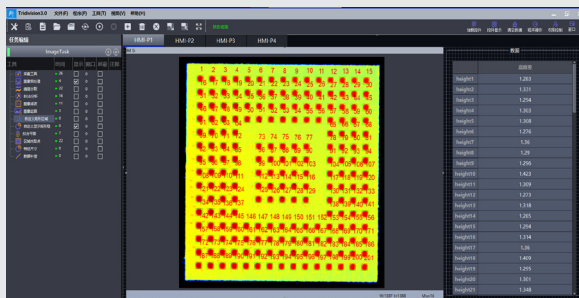
—Automatically generate multiple areas around the solder ball as reference planes for height measurement.



5

Data Compensation → HMI Display

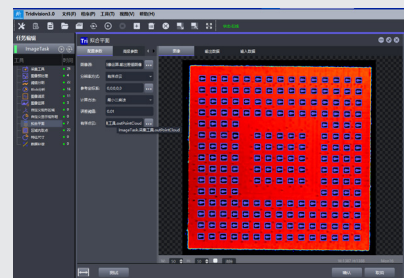
—Apply appropriate data compensation based on product true value references; Real-time observation of the image and measurement area, displaying a list of height measurement data.



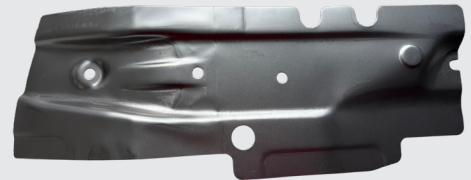
4

Plane Fitting → Point Extraction in Region → Feature Size

—Fit the reference plane, select the highest point of the solder ball, and calculate the height from the point to the surface.



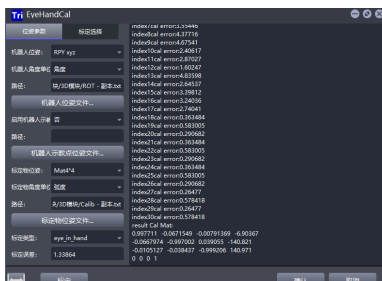
Sample Program—3D - Rivet Hole Guide



1

Hand-eye Calibration

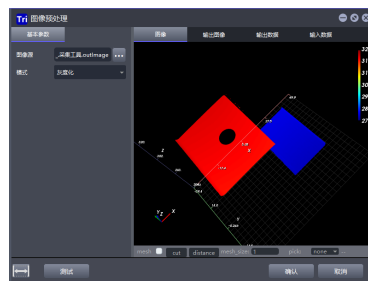
—eye-in-hand calibration,
Generate calibration matrix.



2

Acquisition Tools → Image Preprocessing

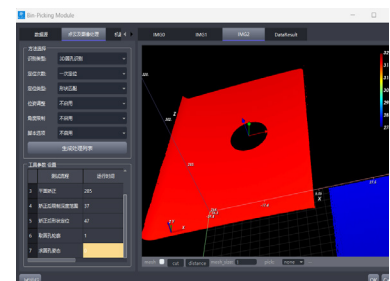
- Connect a 3D camera, acquire the target image, and perform preliminary image preprocessing.



3

Robot 3D Guidance Module

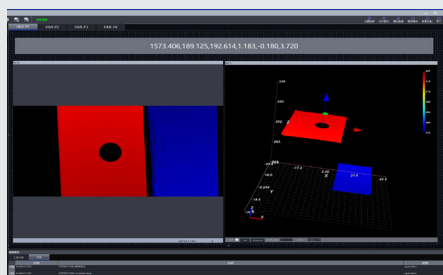
- Extract positioning features based on the "3D Hole Recognition" type and perform pose conversion based on the calibration results.



5

Point Cloud Filtering → Custom Display → Text Conversion

—Set the image display effect and coordinate text display in the HMI interface.



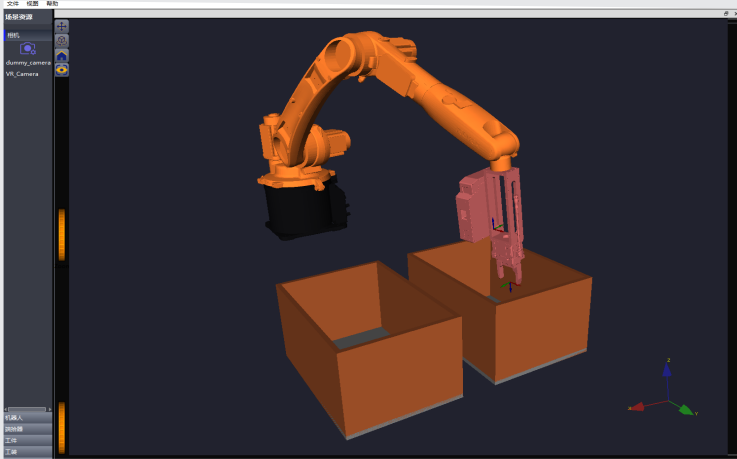
4

Matrix Vector Operations

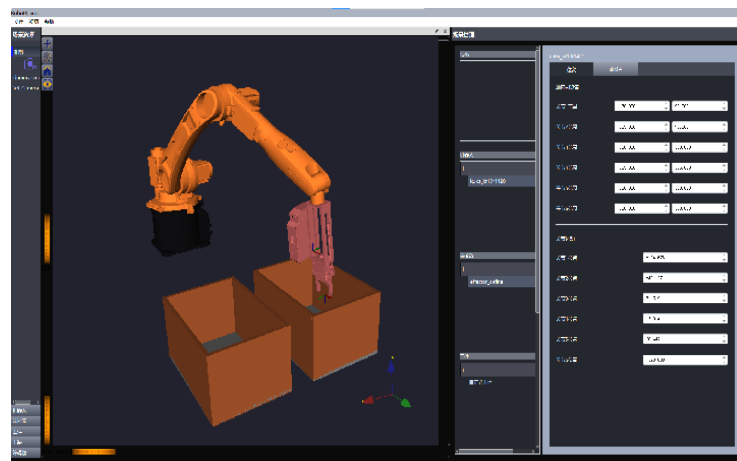
- Perform vector operations and conversions based on the RPY_XYZ operation type, and output the robot coordinate matrix.



Sample Program

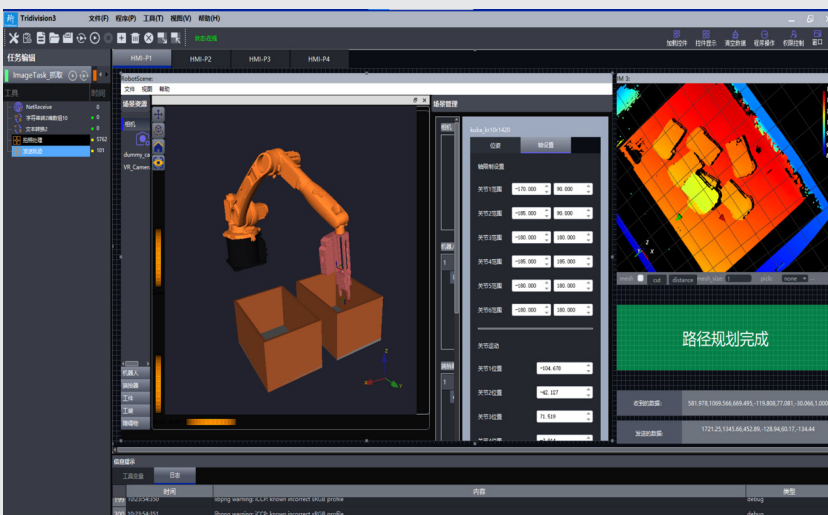


Features of Robot Path Planning



1 Simulation of Real-World Scenarios

—Flexibly add objects within the scene.



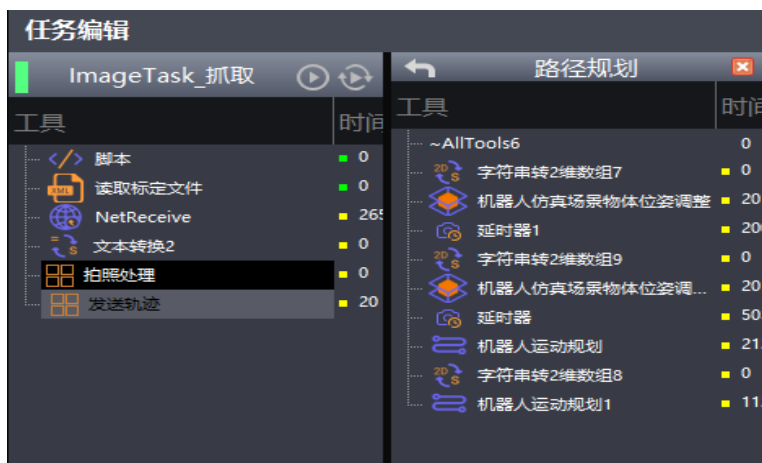
2 Support for Offline Simulation

—Allows for offline observation of the robot's motion trajectory.



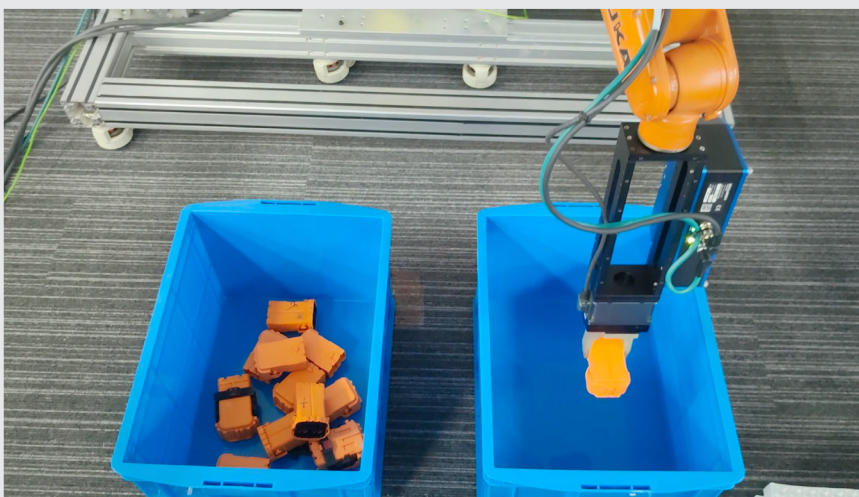
3 Built-in Trajectory Planning and Collision Detection Algorithms

—Plans the optimal path to avoid collisions when targeting grasping positions.



4 Flexible Interaction Between Modules

—Can interact seamlessly with other TridiVision tool modules.

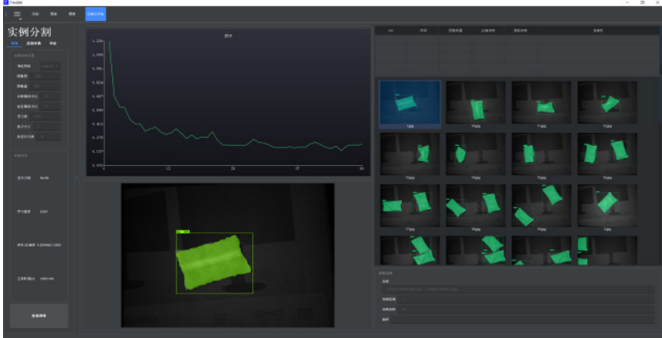


5 Compatibility with Most Mainstream Robots

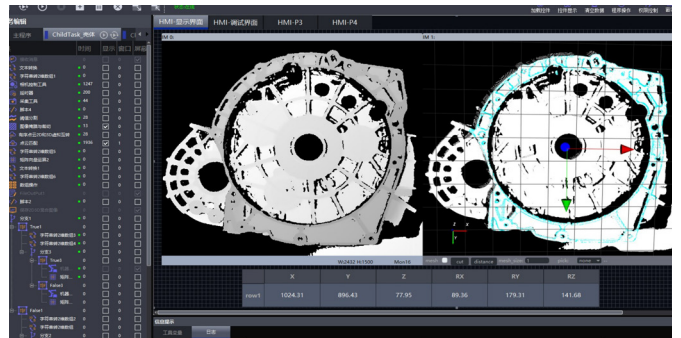
—Communicates with major robot brands through standard interfaces such as TCP/IP.

Demonstration

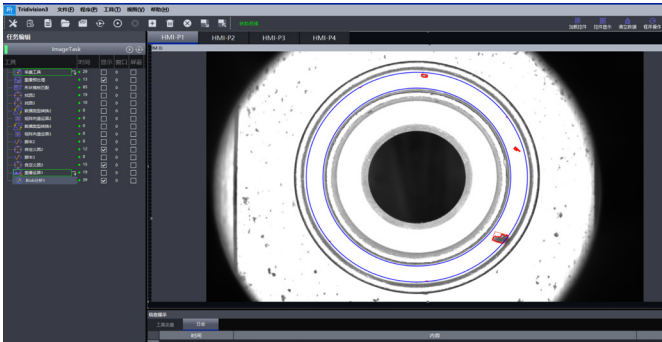
AI-instance Segmentation



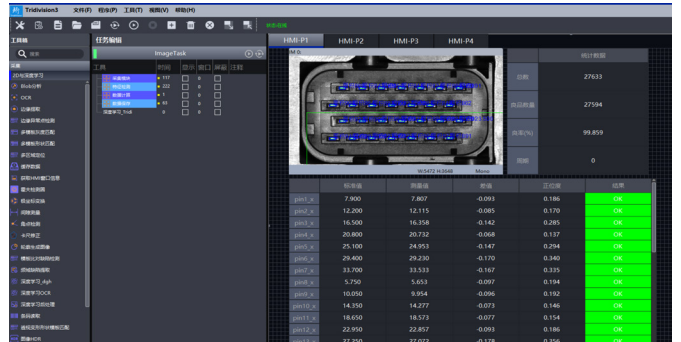
3D-guided Crawling



2D-defect Detection



2D-Position



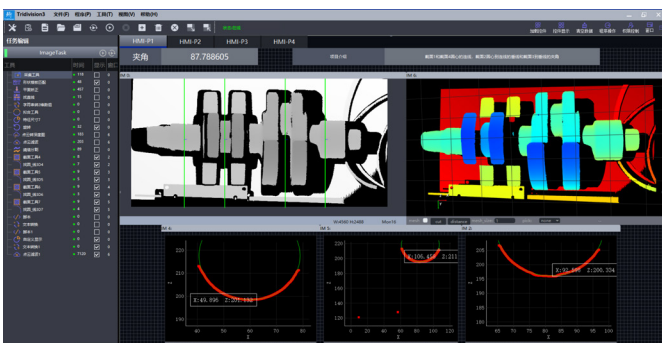
2D-Deep Learning OCR



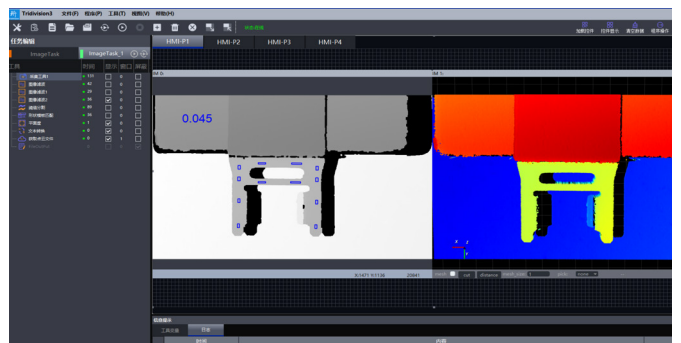
2D-label Detection



3D-Profile Measurement



3D-Flatness



List of Toolkits

Module	Category	List of Main Operators
Acquisition Function Module	Local Point Cloud and Image Data	Acquisition Tool, Save Point Cloud File, Save Image File, Get Point Cloud File, Get Image File, Get Multiple Image Files, Local Disk Operation
	Device and Acquisition Control	Camera Control Tool, Acquisition Tool
	Image-Point Cloud Processing	Point Cloud and Depth Map Conversion - Equal Resolution
2D & Deep Learning	Basic Recognition	OCR, Blob Analysis, Barcode Reading, Matrix Code Reading
	Edge and Shape Detection	Edge Extraction, Edge Abnormal Point Detection, Hough Circle Detection, Polar Coordinate Conversion, Corner Detection, Caliper Correction, Contour to Image Generation, Ring Detection
	Template and Matching	Shape Model Matching, Multi-template Grayscale Matching, Multi-template Shape Matching, Template Comparison Defect Detection, Perspective Distortion Shape Model Matching
	Target Positioning	Multi-region Location, Point Finding Tool, Vertex Finding, Circle Finding, Multi-circle Finding, Line Finding, Multi-line Finding, Ellipse Finding, Rectangle Finding, Region Merging
	Defect and Measurement	Gap Measurement, Frequency Domain Defect Extraction, Pixel Counting
	Basic Image Operation	Image Operation, Image Rotation, Image Preprocessing, Image Filtering, Image HDR, Image Text Writing, Image Window Screenshot, Image Masking and Cropping, Image Quality Evaluation, Image Quality Correction, Get HMI Window Information
	Mosaicking and Fusion	Image Stitching - Based on Known Matrix, Image Fusion
	Segmentation and Color Analysis	Threshold Segmentation, Color Analysis, Color Segmentation
3D	Deep Learning	Deep Learning_Tridi, Deep Learning OCR, Deep Learning Post-processing
	Geometry and Fitting	GeoFeature 3D, Section Cut Tool, Contour Line Type Conversion, Contour Alignment - 3D Line, Holes Pose Fit, Plane Fitting, Flatness, Plane Calibration, Volume Calculation, CircleSingleLine 3D, LineSingle3D, 3D Line Offset, 3D Point-Line Pose Fitting
	Point Cloud Operation	Point Cloud Bounding Box, Point Cloud Comparison, CloudSegment, Point Cloud Expansion, CloudFilter, Point Cloud Registration, SurfaceMatch, Cloud Stitch Based on Known Matrix, Point Extraction in Region, GetPoint_cloud, Image2orderCloud, 3D Point Extraction from 2D in Ordered Point Cloud, GetPointSingleLine, Point Cloud Sequence Operation
	Spatial and Linear Analysis	3D Point Extraction from 2D in Equal Resolution Depth Map, LineIntersect3D, 3D Line Offset, 3D Line Intersection, Perspective Matrix Distortion Calculation, StripeLineExtract 3D, ProfileMatch 3D
	Pose and Positioning	Pose Adjustment, Pose Generation, BoxBagLocation3D, Pose Estimation, Robot Path Planning, 3D Trajectory Generation
Program Logic	Detection	Glue Detection
	Program Control and Logic Operation	HMI Input Parameter Operation, Program Control, CallTask, ReadGlobalVar, Read Message Queue, Send Message, Branch, Task Output, Task Input, Conditional Branch, Loop Break, TimeDelay, CountedLoop, AutoLoadPro, Tool Set, Cache Data Operation, Receive Message, Statistical Data Operation, WriteGlobalVar, Write Message Queue, Array Statistical Operation, License Service, Logic Highlight Setting
Operation & Display	Script Management	Script Tool
	Data Processing and Analysis	Data Compensation, Data Analysis, Data Analysis Operation, Data Check, Data Type Conversion, Data Filtering, Array Operation, FeatureSize, Communication Data Format Conversion, TextCvt, String2Value2DCvt, Coordinate Correction, SaveMixImage, DataComparator, MultySystemCaliper, Foolproof Setting, Change Control State, GetCurrentDateTime, GetToolPara, Get Software Information, Get Array Size, DetectPointCvt, ResultAnalysis, Mat2Vector, LogicMath, Fitting, GetValue, Generate Transformation Matrix, Data Statistics, Array Operation, Position Degree, Line Translation and Rotation, Robot Motion Planning, Robot Simulation Scene Object Pose Adjustment
	Calibration and Correction	Calibration Board Calibration, EyeHandCal, EyeHandPosCvt, Line Laser Calibration, Rotation Center Correction, NPointsCal, NPointsCal3D, NPointCvt, Online N-point Calibration, DistortionRectifying
IO	Custom Display and Region Setting	DefineShowHMI, DefineROI, DefineDlg, DefinePoint, DefineShowROI, DefineShowData, ShowAutoDefine, ShowRect, DefineCircleCaliper, DefineCircle, Custom Line Caliper, DefineSystem
	IO Operation and Communication	CSTControl, CSTConnect, Dasklo Reading, Dasklo Writing, MQTTOperation, MQTTConnect, FyIoRead, FyIo Setting, FyIoSend, IoRead, IO Setting, IoSend, NetReceive, NetSend, NetSet, OpcRead, OpcConnect, OpcWrite, PlcRead, PlcConnect, PLC Data Conversion, PlcWrite, Inovance PLC Reading, Inovance PLC Connection, Inovance PLC Writing, ComConnect, ComReceive, SN Verification, Send Test Result, ComSend, LogWrite, Traceability System Connection
	MES Interface	FTP Operation, HttpRequest, RabbitMQConnect, RabbitMQSend, MES_CATL_CD
Special Function Block	File Operation	FileOutPut, FileRead, WriteXML_FMax, Read Calibration File, ReadWrite.Json, Get File Information, Parse XML Text, Database Operation, Database Connection
	Special Function Block	Point Height Detection Module, Robot 3D Guidance Module, Connector Pin Detection Module, Trajectory Guidance Module

PASSION FOR 3D



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